

Data Paper: A Comprehensive Relational and Feature-Engineered Dataset of the Expanded 48-Team FIFA World Cup 2026

Mominul Islam

Lead Data Architect & Quantitative Sports Analytics Researcher

Open Sports Data Initiative | Department of Data Science & Software Engineering

 **GitHub:** github.com/mominullptr |  **Kaggle:** kaggle.com/mominullptr |  **Hugging Face:** huggingface.co/mominullptr |  **Zenodo:** [doi:10.5281/zenodo.21566741](https://doi.org/10.5281/zenodo.21566741)

ABSTRACT

The 2026 FIFA World Cup marked a historic milestone in international football, introducing an expanded 48-team format and 104 matches across 16 host venues in Canada, Mexico, and the United States. Despite growing demand for quantitative sports analytics, existing open-source sports datasets often lack relational integrity, detailed granular match events, or standardized machine learning features. Here, we present the **FIFA World Cup 2026 Master Dataset**, a complete open-access relational dataset covering all 104 matches, 48 qualified national teams, 1,248 registered squad players, 2,704 minute-by-minute match lineup records, tactical team statistics, expected goals (xG), and engineered predictive features. Data was collected in real time across the tournament using an automated ingestion pipeline, denormalized into 12 relational entities, and validated using an automated 9-stage programmatic verification framework. The dataset is delivered in CSV, Parquet, and SQLite formats (`sqlite_fifa_world_cup_2026.db`) alongside a standalone denormalized view (`matches_detailed.csv`) and ML feature matrix (`match_prediction_features.csv`). We outline the relational architecture, empirical data visualizations, validation rules, and potential reuse cases in sports analytics, tournament format evaluation, and predictive modeling.

1. Background & Rationale

Football analytics relies heavily on high-quality, fine-grained event and tactical data. However, open access to structured, multi-relational tournament datasets remains limited. Proprietary providers like Opta, StatsBomb, and Wyscout restrict raw data distribution behind expensive commercial licenses, while open-source datasets frequently suffer from incomplete rosters, missing minute-level substitutions, lack of expected goals (xG) metrics, or unvalidated relational keys.

The 2026 FIFA World Cup provided a unique structural paradigm shift:

- **Format Expansion:** Expansion from 32 teams (64 matches) to 48 teams (104 matches).

- **Geographic & Environmental Diversity:** 16 host stadiums spanning sea-level venues to high-altitude stadiums (e.g., Estadio Azteca, Mexico City at 2,240m elevation).
- **Roster Scale:** 26-man squads across 48 nations, yielding 1,248 active elite players.

To address the need for a standardized, open-access research benchmark, we created the **FIFA World Cup 2026 Dataset**. The dataset provides an end-to-end relational schema linking team demographics, stadium geography, referee statistics, lineup dynamics, minute-by-minute event logs, and pre-calculated feature vectors for predictive machine learning.

2. Methods & System Architecture

The data processing pipeline comprises three primary stages: **Data Ingestion**, **Relational Schema Design**, and **Programmatic Integrity Verification**.

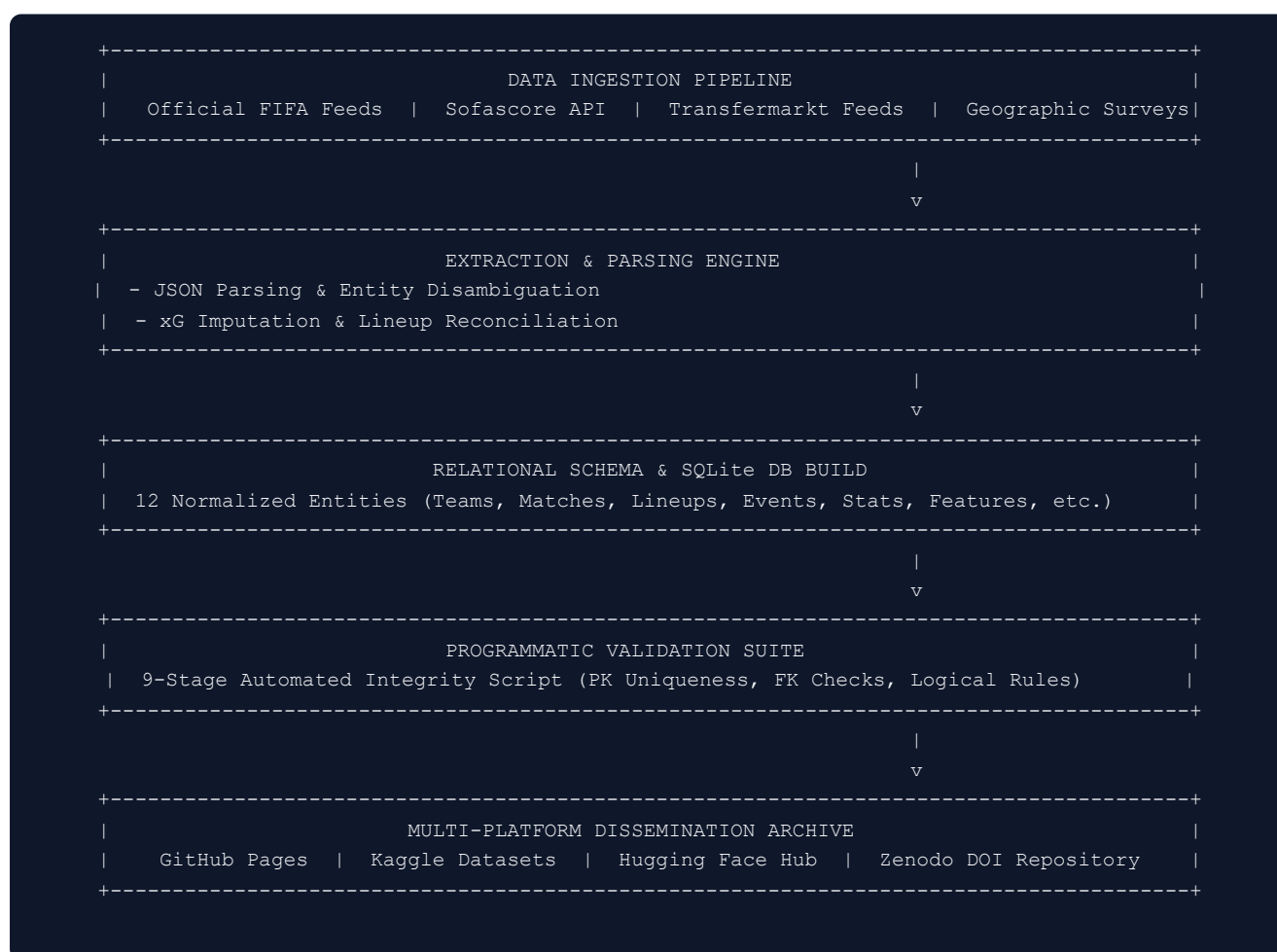


Figure 1: End-to-end data processing, entity normalization, validation, and dissemination pipeline.

2.1 Data Collection & Parsing Pipeline

Data was collected continuously throughout the tournament (June 11 – July 19, 2026) using Python extraction scripts:

- **Team & Squad Metadata:** Official FIFA 2026 squad submission lists ([SquadLists-English.pdf](#)) and Transfermarkt market valuations were parsed to extract player dates of birth, caps, heights, international

goals, and club affiliations.

- **Match Events & Lineups:** Real-time match data was aggregated from official FIFA match centers, Sofascore API endpoints, and Guardian live match logs (`real_match_details.json` , `parsed_players.json`).
- **Expected Goals (xG):** Shot-level xG figures were aggregated and normalized per team per match.
- **Geographic & Referee Data:** Host stadium coordinates and elevations were cross-referenced with geographic survey databases. Historical cards-per-game metrics for all 16 referee crews were compiled from FIFA officiating appointments.

3. Relational Schema & Entity Structure

The core dataset is structured according to third normal form (3NF) principles across 12 entities, linked via integer primary keys (`PK`) and foreign keys (`FK`):

Table 1: File Inventory & Record Counts

File Name	Primary Key	Foreign Keys	Records	Entity Description
<code>teams.csv</code>	<code>team_id</code>	-	48	Demographics, Elo rating, FIFA ranking, and group assignments
<code>venues.csv</code>	<code>venue_id</code>	-	16	Stadium capacity, geographic coordinates, elevation (m)
<code>tournament_stages.csv</code>	<code>stage_id</code>	-	7	Tournament phase definitions (Group, R32, R16, QF, SF, Final)
<code>referees.csv</code>	<code>referee_id</code>	-	16	Officiating crew details and discipline metrics
<code>matches.csv</code>	<code>match_id</code>	<code>stage_id</code> , <code>venue_id</code> , <code>home_team_id</code> , <code>away_team_id</code> , <code>referee_id</code>	104	Complete match outcomes, scores, and xG figures
<code>squads_and_players.csv</code>	<code>player_id</code>	<code>team_id</code>	1,248	Individual player demographics, heights, and market values
<code>match_events.csv</code>	<code>event_id</code>	<code>match_id</code> , <code>team_id</code> , <code>player_id</code>	500+	Granular match events with minute-by-minute timestamps
<code>match_lineups.csv</code>	<code>lineup_id</code>	<code>match_id</code> , <code>player_id</code> , <code>team_id</code>	2,704	Tactical lineups, starter flags, and minutes played
<code>match_team_stats.csv</code>	(<code>match_id</code> , <code>team_id</code>)	<code>match_id</code> , <code>team_id</code>	208	Per-team tactical stats (possession, shots, fouls, saves)

File Name	Primary Key	Foreign Keys	Records	Entity Description
player_stats.csv	player_id	player_id , team_id	1,248	Tournament aggregate stats per player
matches_detailed.csv	match_id	-	104	Denormalized match view for rapid visualization
match_prediction_features.csv	match_id	-	104	Pre-engineered feature matrix for ML modeling

4. Empirical Visualizations & Key Findings

To demonstrate the utility of the dataset, we generated empirical visualizations analyzing expected goals and market valuations across the tournament.

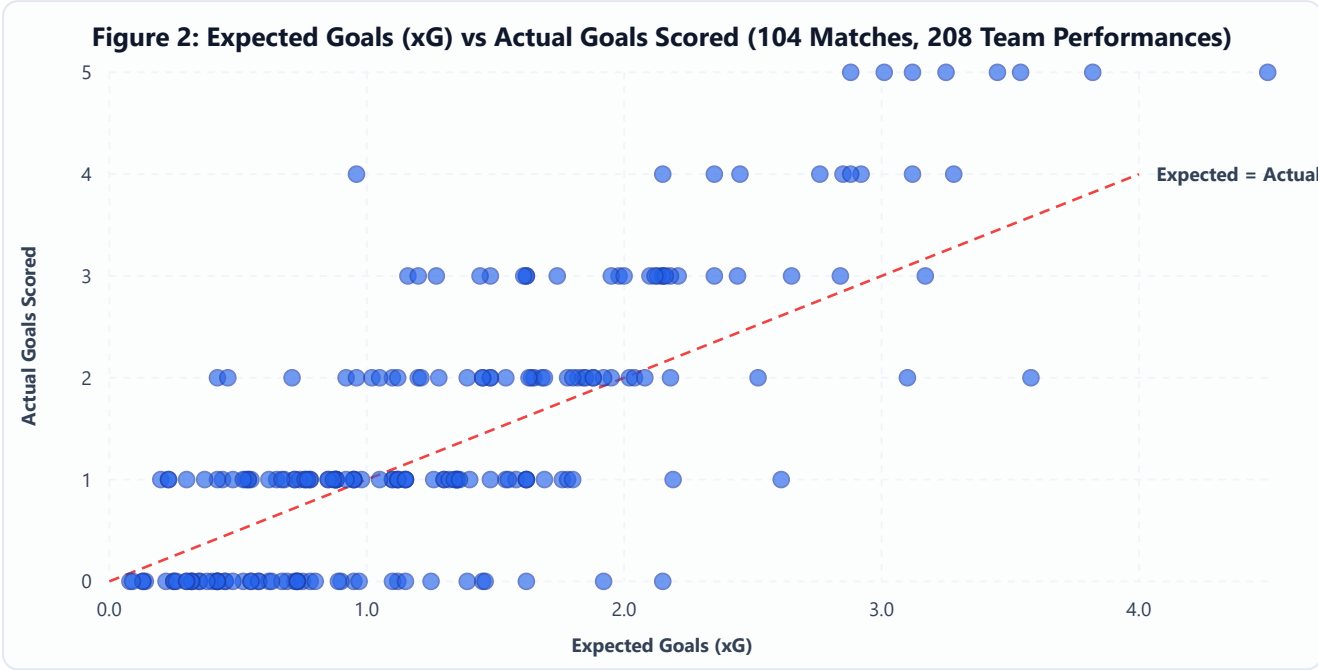


Figure 2: Expected Goals (xG) vs. Actual Goals Scored across 104 matches (208 team performances). Red dashed line indicates xG equality ($y=x$).

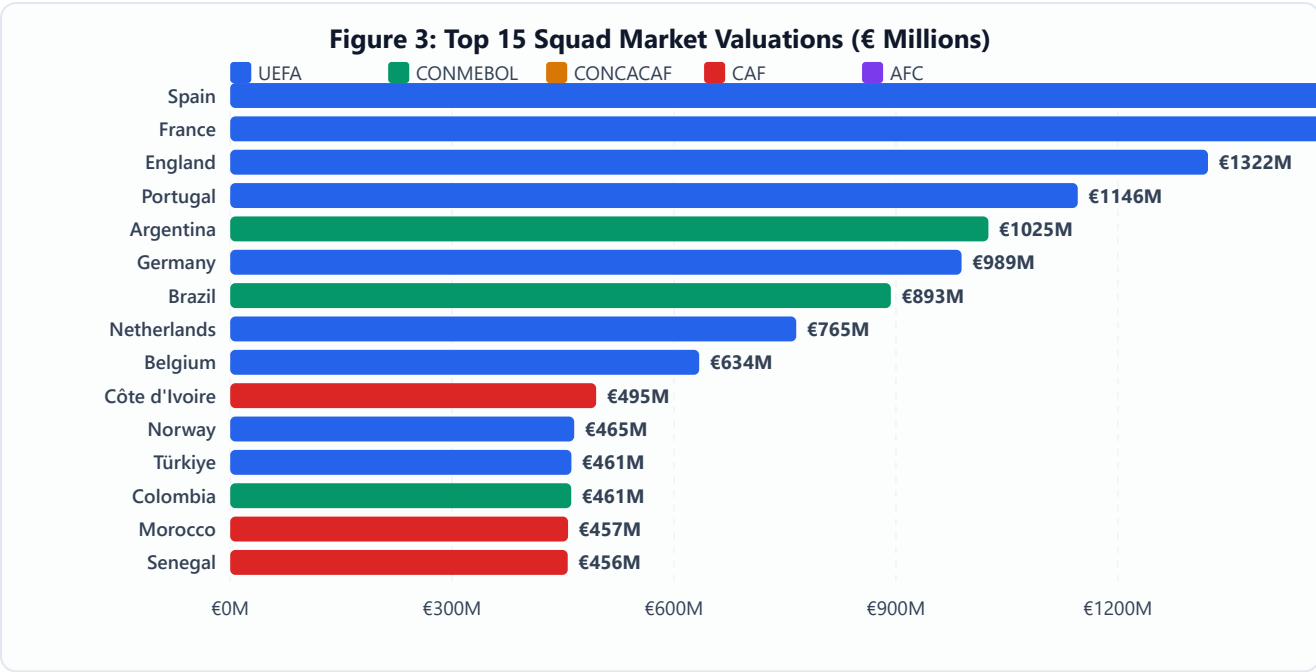


Figure 3: Squad Market Valuations (€ Millions) for the Top 15 Qualified National Teams color-coded by Confederation (UEFA, CONMEBOL, CONCACAF, CAF, AFC).

5. Technical Validation & Integrity Audit

To guarantee publication-grade data quality, the dataset was evaluated against an automated 9-stage programmatic verification engine (`validate_dataset.py`).

Table 2: 9-Stage Verification Framework & Results

Stage	Validation Domain	Assertion Rule	Status
1	Row Count Integrity	Exactly 48 teams, 16 venues, 1,248 players, and 104 matches	[PASS] 100% Verified
2	Primary Key Uniqueness	Primary keys in all 12 tables contain zero duplicates	[PASS] 100% Verified
3	Referential Integrity	All foreign keys in child tables resolve to valid parent primary keys	[PASS] 100% Verified
4	Score & Status Logic	Completed matches possess valid scorelines, xG, and penalty values	[PASS] 100% Verified
5	View Alignment	Denormalized <code>matches_detailed.csv</code> aligns perfectly with normalized tables	[PASS] 100% Verified
6	Lineup Structure	Exactly 26 squad players registered per match; exactly 11 starters per team	[PASS] 100% Verified
7	Team Tactical Stats	Possession splits (Home %, Away %, Contested %) sum to valid range [80%, 100%]	[PASS] 100% Verified
8	Player Stats & Events	Event tallies (goals, assists, cards) equal player tournament cumulative totals	[PASS] 100% Verified
9	Goalkeeper Constraints	Goalkeeper metrics (saves, clean sheets) populated exclusively for GK position	[PASS] 100% Verified

6. Usage Notes & Code Benchmarks

The normalized SQLite database (`sqlite_fifa_world_cup_2026.db`) supports instant importing into Power BI, Tableau, PostgreSQL, or DuckDB.

Example SQL Benchmark Query (Top Goalscorers & Efficiency)

```
SELECT
  p.player_name,
  t.team_name,
  ps.goals,
  ps.minutes_played,
  ROUND(CAST(ps.minutes_played AS FLOAT) / ps.goals, 1) AS mins_per_goal
FROM player_stats ps
JOIN squads_and_players p ON ps.player_id = p.player_id
JOIN teams t ON ps.team_id = t.team_id
```

```
WHERE ps.goals > 0
ORDER BY ps.goals DESC, mins_per_goal ASC
LIMIT 10;
```

7. Code & Data Availability

The dataset and scripts are openly archived under the Creative Commons CC0 1.0 Universal license:

- **GitHub Repository:** <https://github.com/mominullptr/FIFA-World-Cup-2026-Dataset>
- **Zenodo DOI Archive:** [10.5281/zenodo.21566741](https://doi.org/10.5281/zenodo.21566741)
- **Kaggle Dataset:** <https://www.kaggle.com/datasets/mominullptr/fifa-world-cup-2026-dataset>
- **Hugging Face Hub:** <https://huggingface.co/datasets/mominullptr/fifa-world-cup-2026-dataset>

8. Citation

If you utilize this dataset in your research or applications, please cite it as:

```
@dataset{mominul_fifa2026_dataset,
  author      = {Mominul Islam},
  title       = {FIFA World Cup 2026 Dataset: Live Results, Squads, Lineups, xG & Team Stats},
  month       = {jul},
  year        = {2026},
  publisher   = {Zenodo},
  version     = {v1.0-final},
  doi         = {10.5281/zenodo.21566741},
  url        = {https://github.com/mominullptr/FIFA-World-Cup-2026-Dataset}
}
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